CHORD

Cross-cluster Harmonized Optimization of Reception and Dissonance

A bridging, attention-economy feed-ranking algorithm for federated social networks

Working draft

Contents

Abstract . . . 1
1. Motivation . . . 2
2. Design principles . . . 3
3. System overview . . . 3
4. The relation model (keystone) . . . 4
5. Rater weighting: quality-tracking, not variance . . . 5
6. Identifiability and the propensity model . . . 7
7. Value model and feed assembly . . . 8
8. Attention-economy mechanisms . . . 9
9. Estimation and dynamics . . . 10
10. Adversarial robustness . . . 11
11. Federated deployment . . . 12
12. The knob panel . . . 12
13. Limitations and open problems . . . 12
14. Relationship to prior work . . . 14
Appendix A. Notation . . . 15
Appendix B. One-line summary of the architecture . . . 15
Appendix C. Retroactive evaluation . . . 16
Appendix D. Reference architecture for a fediverse deployment . . . 20

Abstract

We describe **CHORD** (Cross-cluster Harmonized Optimization of Reception and Dissonance), a feed-ranking algorithm that inverts the incentive of engagement-based ranking. Where engagement ranking rewards content that maximizes reactions — and therefore structurally favors divisive, high-variance "split-decision" content — CHORD ranks content by a **strength score** that rewards broad support net of divisiveness. The scoring core is built on the bridging-based ranking literature — the matrix-factorization approach deployed in Community Notes and Polis — and takes from the Ethelo collective-decision engine its objective: preferring broadly-supported, low-dissonance outcomes over higher-average but polarizing ones. Around that core we add three attention-economy mechanisms: a **quality-weighted rater model** so that a discriminating reaction counts more than a reflexive

one; a **conserved, strength-replenished author visibility budget** so that high-volume posting dilutes rather than multiplies reach; and a **commons-funded exploration pool** that gives unproven content from newcomers a fair audition. CHORD is structured as a valuation-and-allocation layer between retrieval and presentation, factored into ports so it can run standalone on a single Mastodon instance or dock into a richer host. We ground the estimator's stability in performative-prediction and two-timescale stochastic-approximation theory, correct a rater-weighting pathology identified in deployed bridging systems, and are explicit about what remains genuinely open — chiefly the propensity model on which identifiability rests. A reference implementation is checked against real public datasets (Appendix C) and exercised in a counterfactual agent-based simulator (Appendix C.4); this find-it-then-fix-it loop drove several corrections that are folded into the sections below — an exposure-weighted shrinkage keystone, an outgoing-diversity trust weight and a cluster-spread-gated collusion defense, an anti-bait depth gate, and a bridging-vs-personalization operating point that is interior rather than at the pure-bridging extreme.

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1. Motivation

Recommender systems on social media are best understood not as "content distribution" but as **attention allocation**: from a firehose of candidates they fill a scarce number of attention slots, and in doing so they set the incentives of the attention economy [Ovadya & Thorburn 2023]. Most such systems optimize measurable engagement — clicks, reactions, replies, dwell — which is often a good proxy for what people want but also rewards content that is sensational, outrageous, or divisive. Outrage and ratio-bait are high-*variance* reception events: some love them, some hate them, and that split is what drives the reactions engagement ranking chases.

Two independent lines of work converge on the opposite objective. The **bridging-based ranking** literature, motivated by depolarization, ranks content by whether it earns support *across* a population's divides; it has a governance-tool lineage (Polis) and one deployed instance at scale (X's Community Notes). The **Ethelo** collective-decision engine, motivated by fair group decisions, scores an outcome by average support penalized for **dissonance** (the variance of support across participants) and will rank a lower-average-but-unified outcome above a higher-average-but-polarizing one. These are the same objective reached from two directions, and it is precisely a ranker that penalizes what engagement rewards.

This paper takes that objective as its starting point and builds the ranker around it. The scoring keystone comes from the bridging/collective-response literature (§4); the machinery that makes it identifiable and stable comes from counterfactual learning-to-rank, performative prediction, and two-timescale stochastic approximation (§6, §9); the attention-economy mechanisms draw on provider-side fairness, bandit exploration, and the discoverer literature (§8). Ethelo contributes the objective and the influence-recycling idea of §8. Section 14 gives the full attribution.

This paper assembles the strength objective, the attention-economy mechanisms it implies, and the estimator that makes it identifiable and stable, targeted at the fediverse — where there is no incumbent algorithm, appetite for transparent and swappable ranking, and a governance structure (per-instance, per-user) that makes the necessary value choices legitimate rather than imposed.

2. Design principles

1. **Bridging, not consensus.** Reward support that holds *across* a population's divides. The goal is constructive conflict (conflict *transformation*), not homogeneity or the elimination of disagreement.
2. **Price signal quality, not activity.** A discriminating, cross-cutting reaction is worth more than a reflexive one; an indiscriminate scroller's reaction is near-zero information.
3. **Earn visibility.** The right to be seen at volume is earned through realized broad support, not bought with raw posting.
4. **Circulate influence.** Prevent influence — over rankings or over the model itself — from calcifying into a permanent taste aristocracy.
5. **Fair audition for the unproven.** A newcomer's first exposure is a commons cost, not an author cost, and is decision-theoretically correct, not charity.
6. **Consumption is free; authority is earned.** A user may tune what *they* see without limit; no user may tune how much the system trusts *them*.
7. **Portable and federated.** Users own their preference profile; ranking is transparent, opt-in, and swappable; instances and users set the value parameters.

3. System overview

CHORD is a **valuation-and-allocation layer** sitting between candidate retrieval (upstream, not owned) and presentation (downstream, not owned). It does three things: values attention (which raters count, §5), values content (strength, §4), and allocates scarce visibility (budget + constrained selection, §7–8). Everything else enters through **ports** with crude built-in default adapters (so the core runs standalone) and optional rich adapters (so a capable host upgrades it):

Port	Supplies	Default adapter	Rich adapter
Identity	stable handle + forge-cost	account age/heuristics	verified-human / ZK-pseudonymous
Preference	influent function + history	local store	portable data pod (Solid/ATProto)
Partition	opinion clusters	built-in matrix factorization (§4)	Polis / external bridging service
Signal	attention-event stream	native reactions	richer telemetry
Candidate	posts to re-rank	home/instance timeline	dedicated retrieval
Config	M, ρ, θ, ϵ, constraints	instance defaults	governance UI

The load-bearing observation for federation: identity, preference, signal, and candidate all have *some* native fediverse answer; **opinion-clustering (Partition) does not**, which is why the core must ship its own factorization as the default rather than depending on an external service.

4. The relation model (keystone)

4.1 Factor reception; do not score it separately

The keystone comes from the bridging/collective-response literature: the opinion-embedding factorization below is the Community Notes / Polis approach. Build a **weighted, biased matrix factorization** of the signed reaction matrix r_{up} (e.g. boost +1, favorite +0.5, exposed-no-reaction $-c$, mute -1):

$$\hat{r}_{up} \;=\; \mu \;+\; b_u \;+\; b_{a(p)} \;+\; b_p \;+\; \langle x_u,\, y_p \rangle$$

- $x_u \in \mathbb{R}^d$ — user **opinion embedding** (position in divide-space)
- $y_p \in \mathbb{R}^d$ — post **opinion loading** (which pole it appeals to)
- b_u — rater leniency; $b_{a(p)}$ — **author** baseline; b_p — post intercept

The author term $b_{a(p)}$ is essential and was easy to omit: without it, a famous account's blanket elevation from reach leaks into b_p. With partial pooling ($b_p \sim \mathcal{N}(0, \tau_p^2)$, $b_a \sim \mathcal{N}(0, \tau_a^2)$), b_p measures a post's *marginal* breadth above its author's baseline.

Bridged support and divisiveness fall out as orthogonal quantities. The partisan appeal lives entirely in $\langle x_u, y_p \rangle$; the reception *not* explained by opinion alignment is the intercept b_p. So b_p is a first proxy for bridged support (this is what Community Notes ranks on), and divisiveness is the population spread of the alignment term:

$$D(p) \;=\; y_p^\top A\, y_p$$

With whitened embeddings and $A = I$ this is just $\|y_p\|^2$ — universal posts sit near the origin of opinion space; partisan posts sit far out. But $A = I$ is too glib: it penalizes a "fishing vs. libraries" split as much as a political fault line. We take $A \succeq 0$ **weighted toward the axes that correlate with an affective-polarization signal** (regress the signal on embedding dimensions). This is where the instance's "which divides do we care to bridge" decision physically lives (the ρ knob, §7).

4.2 Do not trust the scalar intercept — reconstruct per cluster

The scalar b_p is a linear proxy for diverse approval and diverges from the real target when clusters sit asymmetrically about the origin (it happily rewards a post one cluster loves and another merely tolerates). Using the Partition port's clusters c, reconstruct each cluster's predicted reception

$$\hat{r}_{cp} \;=\; \mu + \bar{b}_c + b_{a(p)} + b_p + \langle \bar{x}_c,\, y_p \rangle$$

and define bridged support by **shrinking each cluster's reception toward the population mean by how much that cluster was actually exposed, then aggregating across clusters**:

$$\boxed{B_{\mathrm{LCB}}(p) \;=\; \operatorname*{agg}_{c}\; \tilde{r}_{cp}, \qquad \tilde{r}_{cp} \;=\; \bar{r}_p \;+\; \frac{n_{cp}}{n_{cp} + n_0}\,(\hat{r}_{cp} - \bar{r}_p)}$$

where $\bar{r}_p = \frac{1}{|C|}\sum_c \hat{r}_{cp}$ is the population (cluster-agnostic) reception and n_{cp} is the (propensity-corrected, §6) number of cluster-c users actually *exposed* to p. This is an empirical-Bayes (James–Stein / DerSimonian–Laird) shrinkage: a cluster that has barely been exposed ($n_{cp} \to 0$) regresses to the mean — its apparent dissent is sampling noise, not a tested divide — while a **well-exposed** disagreeing cluster keeps its low reception and pulls the score down. **A post is not credited as bridging above the mean until it has survived contact with the people who would dislike it.** The aggregator is a knob: the default agg = nash (geometric mean of per-cluster agree-probabilities) is Polis's group-informed consensus — one opposed group blocks high consensus, without the brittleness of a hard min; agg = min recovers Ethelo's Rawlsian worst-cluster, and an Atkinson equally-distributed-equivalent interpolates between them. Keep the scalar b_p as a cheap pre-filter; rank on B_{LCB}.

Note the deliberate asymmetry in how uncertainty is used: the exploration pool (§8) samples *high*-uncertainty posts optimistically to decide what to **audition**; B_{LCB} shrinks *low*-exposure clusters to the mean and credits only tested breadth to decide what to **crown**. Optimism explores; conservatism rewards. This is what prevents imperfect estimates from manufacturing false bridging.

This form was arrived at empirically, against a deployed baseline (Appendix C.5). An earlier version used a subtractive penalty $B_{\mathrm{LCB}} = \min_c[\hat{r}_{cp} - \beta\sigma/\sqrt{n_{cp}+1}]$. Benchmarked against X's Community Notes — the one deployed bridging system whose helpful/not decisions we can treat as ground truth — that version was *beaten by both* the scalar b_p and a naive mean of signed helpfulness at recovering CN's `CURRENTLY_RATED_HELPFUL` status, because with a rating-count n_{cp} the min-minus-penalty demotes notes *sampled* by fewer clusters (noise) rather than notes *divisive* across clusters (risk) — it subtracted noise, not risk. Two changes fixed it and are now the shipped default: replace the subtractive penalty with the exposure-weighted shrinkage above, and aggregate with the nash product rather than a hard min. On a class-balanced Community Notes sample the repaired B_{LCB} reaches b_p parity — it can no longer be beaten by its own intercept — and on Polis, where genuine multi-group structure makes bridging differ from the mean, it tracks true cross-group support far better (Spearman 0.80 → 0.97). The one standing requirement is that n_{cp} be a real propensity-corrected *exposure* count (§6), not a raw rating count; against CN's own intercept-thresholded label the naive mean remains unbeatable by construction, so the decisive evidence for the aggregator is Polis, not CN.

5. Rater weighting: quality-tracking, not variance

Each observation in the fit is weighted by a per-rater influence λ_u, so that discriminating cross-cutting raters dominate the estimate and indiscriminate scrollers barely move it ("price selectivity, not activity"). **The naive choice is actively harmful.** Inverse-variance weighting rewards *predictability*: extremists are predictable, so they get low residual variance and high weight, while thoughtful evaluators get inflated residuals and are downweighted — empirically, this amplifies ideologically extreme raters and makes the system *more* vulnerable to partisan attack [Quality-Sensitive MF for Community Notes, 2026].

The correction is a **quality-tracking / peer-prediction weight**, estimated jointly inside the factorization: give more influence to raters whose *ideology-adjusted* ratings are consistent with the note-quality estimate learned from all ratings. This is also the modern form of the recursive cross-divide credibility used by YourView ("you are credible if trusted by credible people who disagree with you"), computable via a damped, teleporting eigentrust iteration on the learned geometry:

$$\lambda \leftarrow \tfrac{1-\delta}{n}\mathbf{1} + \delta\, T^{\top}(w\odot\lambda), \qquad w_v = \frac{H(T_{v\cdot})}{\log \deg^{+}(v)}, \qquad T_{vu} = \frac{\sum_{p:\,a(p)=u} [r_{vp}]_{+} \cdot \operatorname{dist}(x_v, x_u)}{\sum_{u'} \sum_{p:\,a(p)=u'} [r_{vp}]_{+} \cdot \operatorname{dist}(x_v, x_{u'})}$$

where $w_v \in [0,1]$ is the **outgoing-diversity** weight — the normalized Shannon entropy of rater v's trust row — introduced below to defeat collusion rings; $w_v = 1$ recovers plain eigentrust.

T must be **row-stochastic** — normalized over each *rater's outgoing* trust ($\sum_u T_{vu} = 1$), not over each author's incoming — so a rater distributes one fixed unit of trust among the authors it approves (classic EigenTrust). The choice is load-bearing for Sybil starvation: under *column* normalization a Sybil author boosted by a single colluding puppet would inherit that puppet's entire weight (its lone incoming edge normalizes to 1), whereas under row normalization an honestly-approved author accrues from *many* independent cross-divide raters while a one-puppet Sybil receives only that puppet's fraction. The teleport floor ($\delta < 1$) makes this a contraction with a unique fixed point and floors every rater's own weight (no one is zeroed). But the floor is also an attack surface:

Row-stochasticity alone is not enough — a *ring* beats it, so we add outgoing-diversity weighting (Appendix C.5). Row normalization defeats the *one*-puppet attack but not a coordinated ring, as validation on real signed votes (Wikipedia RfA) made concrete. The teleport floor gives *every* account — puppets included — a baseline weight of $(1-\delta)/n$. A ring of K fresh accounts that each cast exactly one approval, all of it aimed at one target author, would hand that target $\delta \sum_i \lambda_{\text{puppet}_i} \approx \delta K(1-\delta)/n$ of transported mass: each puppet, being row-stochastic, forwards its *entire* unit to the target, while honest boosters split their unit across the many authors they genuinely approve. The puppets themselves stay starved, but the *target* harvests their pooled baseline mass — the original ranker let the target climb from the 75th percentile of real-editor influence at $K = 5$ to the 100th at $K = 100$. This is not a CHORD-specific bug: no symmetric reputation function is Sybil-proof (Cheng & Friedman 2005), and the uniform teleport $\frac{1-\delta}{n}\mathbf{1}$ is exactly that symmetric part.

The fix, now the shipped default, is the transmit weight w_v above: weight each rater's *outgoing* trust by the normalized entropy of its trust row, so a **single-target rater — every ring puppet, out-degree 1 — transmits** $w_v = 0$, forwarding none of its floor mass, while an honest rater who spreads approval across many authors keeps $w_v \approx 1$. On RfA this collapses the ring target from the 100th percentile back to the 0th at every K, at *no* cost to honest ranking (the influence order of real editors is unchanged, Spearman 1.0), because the attack's signature — concentrating one's entire outgoing unit on a single beneficiary — is precisely what the entropy weight penalizes; an attacker who dilutes a puppet across several authors to raise its entropy thereby splits its mass *away* from the target. A small floor on w_v keeps a genuine low-activity newcomer from being muted. This is a first line, not a completeness proof: the theoretically complete defense is an *asymmetric* teleport (a pre-trusted seed set à la TrustRank / EigenTrust's pre-trusted peers), which CHORD supports as an optional seeded restart bound to the identity port (§11); combined with the identity forge-cost it closes the residual. **Whichever estimator is used, weight by agreement with the bridged-quality signal after ideology is projected out — never by residual variance.**

Two earned quantities feed off the same geometry:

- **Scout precision** $q_{\text{scout}}(u)$ — reward being *early* on posts that *eventually* score high strength. Self-correcting, because it is graded against future outcomes, not current consensus:

$$q_{\text{scout}}(u) \;=\; \frac{\sum_{p\in P_u^+} e^{-\alpha\,\text{rank}_t(u,p)}\,\Phi_\infty(p)}{\sum_{p\in P_u^+} e^{-\alpha\,\text{rank}_t(u,p)}}$$

- **Consumption vs. authority.** A user's *consumption* appetite along any factor is a free knob; their *authority* (λ, scout precision) is earned. Declaring scout-inclination can route more auditions to a user, but their verdict carries scout-weight only if their record earned it.

6. Identifiability and the propensity model

6.1 The problem

Reactions are observed only where a user was *shown* a post, and every historical policy shows people mostly in-group content. So a divisive post can look bridging simply because the outgroup that would hate it never saw it — the negatives are **missing, not negative**, and the missingness depends on the very alignment we are estimating (MNAR). Uncorrected, this inflates b_p and shrinks $\|y_p\|$: in-group popularity masquerades as bridging.

6.2 The correction

This is the counterfactual learning-to-rank problem. Weight each observation by inverse propensity $1/\pi_{up}$ (the probability the pair was exposed), which yields an unbiased objective **provided propensities are accurate and non-zero for every relevant pair** [Joachims, Swaminathan & Schnabel 2017; Schnabel et al. 2016]. The non-zero-everywhere (positivity) requirement is exactly why the exploration pool is not optional — it guarantees $\pi \geq \epsilon > 0$ for all items, and its known, alignment-*independent* exposure is the **unconfounded anchor** that makes organic propensities estimable and checkable. Use the self-normalized estimator to control IPW variance, and treat silent disagreement (exposed-but-no-reaction) as a weak negative so haters scrolling past stop reading as "missing."

$$\mathcal{L}(\Theta) \;=\; \frac{\sum_{(u,p)\in E} \omega_{up}\,(r_{up} - \hat{r}_{up})^2}{\sum_{(u,p)\in E} \omega_{up}} \;+\; \Omega(\Theta), \qquad \omega_{up} \;=\; \lambda_u \cdot \min\!\Big(\tfrac{1}{\hat{\pi}_{up}},\, W_{\max}\Big) \cdot s_{up}$$

Tie the clip to the exploration floor: with $\pi \geq \epsilon$ guaranteed for audited items, setting $W_{\max} = 1/\epsilon$ gives a natural ceiling on inverse weights and hence on gradient variance. This is a variance-for-bias trade, not a free structural guarantee — a genuinely in-group-misaligned pair (an outgroup member who would dislike a post and was never shown it) can have true per-pair propensity *below* ϵ, so clipping there biases exactly the deep-MNAR pairs the correction exists to recover. Adopt the cap, but size it knowing it slightly under-corrects the hardest cases.

Weight scale vs. regularization. The data term is self-normalized (divided by $\sum \omega$) and hence invariant to a global rescaling of ω, but $\Omega(\Theta)$ is *not*. Because λ_u arrives as a normalized distribution ($\sum_u \lambda_u = 1$ from the §5 eigentrust fixed point), the raw ω_{up} are $O(1/|E|)$, so a fixed Ω silently dominates the fit and collapses the embeddings toward the origin. Rescale ω to unit mean over E before each solve (equivalently, read Ω's strength relative to mean-one weights); by the self-normalization above this leaves every estimate unchanged while restoring a well-conditioned regularized problem.

6.3 The propensity model — an open menu, not a commitment

Because misspecified propensities silently reintroduce the MNAR bias the whole edifice corrects, we deliberately keep this pluggable and prefer estimators that fail gracefully:

- **(a) Doubly-robust (default posture).** Pair *any* propensity estimate with a reception imputation model; the estimator is consistent if *either* the propensity *or* the imputation is right [Dudík et al. 2014; Saito 2020]. This is the safest default and should wrap whatever else is chosen.
- **(b) Exposure/position click models** (cascade, examination) — classic CLTR propensity from the logging policy's presentation.
- **(c) Policy-derived propensity** — if the logging ranker's scores are known, derive $\hat{\pi}$ from its (softmax) selection probabilities directly.
- **(d) Intervention harvesting / randomization** — use the exploration pool's known $\pi \approx \epsilon$ as ground-truth anchor to calibrate or validate (b)/(c).
- **(e) Propensity-light fallbacks** — affine corrections or propensity-independent bias recovery for regimes where propensities are unreliable [Vardasbi et al. 2020].

The throughline across all options: **the exploration pool is the randomized, unconfounded mass that anchors identifiability**, so its rate ϵ can never be driven to zero.

7. Value model and feed assembly

7.1 Personalized value

$$V(u,p) \;=\; \underbrace{B_{\mathrm{LCB}}(p)}_{\text{tested bridged support}} \;+\; \underbrace{(1-M)\,\langle x_u, y_p\rangle}_{\text{personalization}} \;-\; \underbrace{M\,\rho\,y_p^{\top} A\, y_p}_{\text{divisiveness penalty}}$$

$M \in [0,1]$ is the master dial. $M = 0$: give me what my side likes, divisiveness included (engagement-like). $M = 1$: pure bridging — broad tested support only, partisan lean ignored, divisiveness penalized. This is a **consumption** choice and therefore ungameable — it only changes the chooser's own feed. Note $M = 1$ is *not* the welfare-optimal setting: in the closed-loop simulator (Appendix C.4) $M = 1$ is **Pareto-dominated by an interior** $M \approx 0.7$, which delivers more genuine (quality $\times$ bridged) value *and* less divisiveness *and* no loss of satisfaction — keeping a little personalization surfaces content that is both liked and actually good, whereas the pure-bridging corner over-corrects. The default sits near this knee, not at the extreme.

Here A is the *fixed* ($\rho = 1$) divide-weighting of §4.1 and $D(p) = y_p^{\top} A\, y_p$ is likewise defined at $\rho = 1$ (Appendix A); the ρ knob enters the value **exactly once**, as the coefficient on the penalty term. The §12 shorthand "ρ scales A" is realized *here* — do not additionally fold ρ into A inside D, or the knob is applied twice.

7.2 Feed as constrained selection

Per viewer, choose N slots to maximize value under constraints. This follows Ethelo's framing of a decision as picking the combination that maximizes strength subject to constraints; the feed realizes it as submodular top-k re-ranking with a greedy $1 - 1/e$ guarantee:

$$\max_{S:\,|S|=N} \sum_{p \in S} \Big[(1-\epsilon)\ \textstyle\sum_f \theta_f V_f(u,p)\ +\ \epsilon\,\tilde{\Phi}(p)\Big] \cdot \mathrm{posdisc}(p)$$

subject to: per-author cap and budget ($\sum_{p \in S,\, a(p)=a} E(p) \leq B(a)$, §8); **diverse-*approval* coverage** (submodular — diminishing returns for re-covering an already-covered region of opinion space); exploration floor $\geq \epsilon N$. Greedy gives the $1 - 1/e$ guarantee; run it over the top few hundred candidates, not the corpus. Meet the hard floor by rounding **up**, $\lceil \epsilon N \rceil$: on the small feeds typical of a fediverse instance a floor of $\lfloor \epsilon N \rfloor$ rounds a sub-unit reservation to zero and silently forfeits the positivity guarantee $\pi \geq \epsilon > 0$ that §6.2 identifiability rests on. (The *randomized* identifiability anchor of §6.2 — the slice whose unbiasedness needs the exposure rate to equal ϵ **in expectation**, not merely to be nonzero — is instead realized by *stochastic* rounding of ϵN: reserve $\lfloor \epsilon N \rfloor$ and one more with probability $\epsilon N - \lfloor \epsilon N \rfloor$.)

> **Diverse approval, not diverse exposure.** The intuitive "show people the outgroup" move has poor validity as a bridging measure — evidence suggests indiscriminate outgroup exposure can *increase* affective polarization [Ovadya & Thorburn 2023, §Evaluation]. The constraint rewards content that *earns* cross-cluster support, never forced exposure.

7.3 The factor vector

V_f are per-factor value functions (trend, scout, depth, locality, recency…), each with its own per-rater precision, its own consumption weight θ_f (a free user knob on the simplex), and its own slider. Trend and scout are latency-coupled: scout-strength is a leading indicator of trend-strength, so users cannot fully opt out of the scout factor without starving future discovery (a commons argument for the exploration pool).

8. Attention-economy mechanisms

Author visibility budget (conserved + earned). Per author, per window, exposure is capped and replenished by realized strength:

$$\sum_{p:\,a(p)=a,\,p \in W} E(p) \leq B(a), \qquad B_{t+1}(a) = \mathrm{clip}\Big(B_0 + \eta \sum_{p:\,a(p)=a,\,p \in W_t} [\Phi(p)]_+\, E(p),\ \ 0,\ B_{\max}\Big)$$

Two guards keep the replenishment well-behaved: the rectifier $[\Phi(p)]_+$ makes a net-divisive post ($\Phi < 0$) simply *fail to replenish* rather than draining the author's floor B_0 (which would double-punish and could drive $B(a)$ negative), and the clip to $[0, B_{\max}]$ keeps every budget bounded — a precondition the §9 bounded-regime argument relies on. Firehose posting spreads a fixed budget thin; quality regenerates it. This inverts the engagement logic (where each post is an independent virality lottery ticket, so volume is rational) into one where posting more *dilutes you* unless it earns. The budget binds to the **identity** port, not the raw account, so it cannot be sharded across sockpuppets.

Exploration pool (cold-start, base-rate-calibrated). New posts have no b_p. Audition them via Thompson sampling — but **not** with a flat optimistic prior, which systematically over-explores weak items because the real base rate of "winners" is far below 50% [Dynamic Prior Thompson Sampling 2025]. Initialize the prior to the empirical newcomer strength rate. A commons-funded fraction ϵ of every feed's slots is reserved for high-uncertainty posts, routed preferentially to **high-q_{scout} raters** (they resolve uncertainty in the fewest impressions). The audition closes on **evaluation saturation** ($\hat{\sigma}_p^2$ below threshold), not wall-clock, so slow-burn long-form is not buried. Grants are bounded **per verified identity** (a periodic audition for humans; nothing for Sybil farms).

Influence recycling (anti-ossification). Damp the consistently-satisfied, boost the under-served, so the system listens hardest to whoever it serves worst:

$$\lambda_u^{\text{eff}} \;=\; \lambda_u \left(1 + \zeta \,(\overline{S} - \bar{S}(u))\right)$$

with $\bar{S}(u)$ the user's model-estimated realized value over what they were shown (hard to fake by "acting dissatisfied"). This is the governor that keeps §5's rater-weighting from calcifying into a taste aristocracy — the one mechanism most ranking systems lack.

9. Estimation and dynamics

9.1 The loop

Per window: **(1)** fit the doubly-robust, IPW-corrected, λ-weighted MF (block-convex ALS, closed-form per block) → embeddings, biases; **(2)** whiten, recompute A and D; **(3)** update quality-tracking λ on the new geometry; **iterate 1–3**; **(4)** update q_{scout}; **(5)** update author budgets; **(6)** update Thompson posteriors; **(7)** apply recycling → λ^{eff}. Per request: retrieve candidates → score $V(u, p)$ with the user's knobs → greedy constrained select → serve → log.

9.2 Why it is stable — and the exact conditions

Two coupled instabilities, each with a governing theory:

- **The λ↔x coupling** (credibility depends on geometry; geometry is fit weighted by credibility) is structurally an **actor-critic**, hence **two-timescale stochastic approximation** [Borkar 1997]: the fast "critic" is the embedding fit tracking a quasi-static λ; the slow "actor" is the λ update. Almost-sure convergence holds under timescale separation (fast step / slow step → 0) plus a stability condition [Lakshminarayanan & Bhatnagar 2017], with concentration bounds [Borkar & Pattathil 2018] and finite-time rates [Doan 2023]. The theorem needs each timescale's limiting ODE to have a unique stable equilibrium; the fast one (embedding given λ) has this **only if strongly convex**, which the exploration anchor provides. So the anchor is the *precondition* that makes the convergence result apply — not a separate trick.
- **The outer allocation→data loop** (rankings change the reactions we then train on) is **performative prediction** [Perdomo et al. 2020]. Repeated retraining converges linearly to a **performatively stable** point iff performative effects are weak and well-conditioned (roughly, distributional sensitivity below the strong-convexity / smoothness ratio). Above that threshold it can oscillate or diverge. This gives a concrete safety target: keep the sensitivity of the reaction distribution to a re-rank small relative to loss curvature — mechanically bought by the exploration anchor and by slow knob changes. The λ-memory case is the *stateful*

variant [Brown, Hod & Kalemaj].

9.3 Stability as a monitored runtime property

Because global convergence is **not** guaranteed in the nonconvex regime, the right target is a **bounded stationary regime**, held by: persistent excitation ($\epsilon \geq \epsilon_{\min}$, so the system never stops sampling regions it stopped showing); slow knob changes; SNIPW + clipping to bound gradient variance; under-relaxation and two-timescale separation. Run a **controller on the estimator's own concentration**: track effective rater count $(\sum \lambda)^2 / \sum \lambda^2$ (or Gini(λ)); if it collapses, automatically raise the teleport floor δ and the damping. The exploration pool is therefore load-bearing four times over — provider fairness, cold-start, causal identification, and estimator stability — and its rate is a floored system invariant, not a user preference.

Separating endogenous from exogenous shift. The reaction distribution moves for two reasons: the ranker's own allocations (endogenous, a feedback loop to *damp*) and real-world drift such as breaking news (exogenous, a signal to *track*). Damping both chases the loop *and* lags real events; tracking both amplifies the loop. The exploration floor already supplies the instrument to tell them apart: the non-personalized slice (chronological or randomly-ranked exposures at rate $\geq \epsilon_{\min}$) is not driven by the personalized ranker, so distributional drift measured *there* is an estimate of exogenous background shift, while drift in the personalized stream *beyond* that baseline is attributable to the loop. Feed the exogenous estimate to the controller as an offset — raise damping and knob inertia only in response to the residual endogenous component — so the system holds steady against its own feedback without treating a genuine news event as instability to suppress. This reuses machinery already present (the exploration cohort) rather than adding a subsystem.

10. Adversarial robustness

- **Sybil / sockpuppets** — the visibility budget binds to identity, not accounts, and trust propagation gives fresh accounts ≈ zero weight *as raters*. A naive teleport-floor eigentrust had a residual hole — a ring of fresh accounts all approving one target let that *target* harvest the puppets' floor mass — which the **outgoing-diversity transmit weight** now closes: a single-target puppet forwards zero trust, so the ring collapses to the floor at any size (§5, Appendix C.5, validated on RfA). Sharding across accounts therefore gains nothing, for puppets *or* their beneficiary; the optional seeded-teleport asymmetry plus the identity port's forge-cost (§11) close the theoretical remainder. **Residual (found in the simulator, App C.4):** a *distributed* ring that camouflages puppets across clusters — each rating genuine content to embed in a real cluster, then boosting one target — manufactures fake cross-cluster support that defeats the B_{LCB} min *and* is invisible to the out-diversity weight (the puppets are not single-target *raters*). A plain co-approval discount only partially contains it, but a **cluster-spread-gated loyalty discount** does — it keys on the one act camouflage can't hide (the same accounts approving *every* target post over time) and gates by the bloc's opinion-cluster spread, driving the ring's amplification below a legitimate author's; the exploration-anchor cap and spectral spike-removal are the principled hardenings still open (§13.10).
- **Brigading** — two independent defenses: a brigade of fresh accounts has no cross-divide trust path, and a brigade *creates* a split distribution that the divisiveness term and the B_{LCB} min-over-clusters penalize. Gaming lowers the score.

- **Bridging-bait (Goodhart)** — shallow universal content (cat memes) can score high bridged support. The depth handling now resists it structurally: a per-post depth/quality signal both *rewards* genuine depth and multiplicatively **gates** a shallow post's positive bridged support toward a floor, so breadth alone cannot buy a crown. In the simulator this roughly halved the gap to an oracle and cut a bait author's reach ~6× (§4.2, App C.4). It reduces rather than eliminates the failure — a baiter who can forge a high depth *signal* still profits, so the signal's own integrity matters.
- **High-precision collusive clique** — genuinely the hardest residual: colluding experts read like independently-delighted experts, and peer-prediction weighting has equilibria where raters agree with *each other* rather than with truth. Only timing/provenance separates collusion from consensus. Named, not solved.

11. Federated deployment

Every port ships a crude default so the core runs on a single instance, and a rich-adapter slot so a capable host upgrades it. On Bluesky's AT Protocol, pluggable "feed generators" make the whole layer first-class; on Mastodon/ActivityPub there is no equivalent standard, so deployment is per-instance or client-side. A maximal host (e.g. a Polity-class stack) fills every port — verified/ZK identity, portable data pods, Polis clustering, a feed substrate, governance UI — and thereby serves as an existence proof that all six ports can be filled; but the design depends on *none* of it, and treats such a host as one adapter set among many. Two natural levels: an instance runs a strength-ranked community timeline (the collective-surface use the bridging/collective-response lineage fits most naturally); individuals get personalized ranking on top, with M, ρ, θ, ϵ as their own dials.

12. The knob panel

Knob	Meaning	Range	Who sets it
M	bridging vs. personalization	$[0, 1]$	user (free)
ρ	collective-identity / which divides to bridge	$[0, 1]$, scales A	instance default + user override
θ_f	factor mix (trend/scout/depth/locality…)	simplex	user (free)
ϵ	exploration appetite (floored)	$[\epsilon_{\text{min}}, \epsilon_{\text{max}}]$	user (free), floor system-set
$\lambda_u,\ q_{\text{scout}},\ B(a)$	rater/author authority	—	**earned, never user-set**

The last row is the consumption-vs-authority wall, enforced structurally.

13. Limitations and open problems

Honest residuals, several of which no fix fully removes:

1. **Conditional convergence.** All guarantees assume *weak* performativity and, for the inner loop, a locally unique equilibrium the anchor only provides *locally.* Under strong performative effects the system can oscillate; we monitor rather than prove.
2. **Unobserved confounding in the propensity model** (§6) is the softest load-bearing wall — and the residual is *bias*, not variance. The variance failure (weights exploding as $\pi \to 0$) is handled by SNIPW, doubly-robust estimation, the $1/\epsilon$ clip, and the B_{LCB} pessimism. What none of these touch is a hidden variable that drives *both* exposure and reaction but is absent from the model: no calibration, clipping, or doubly-robust wrapping removes confounding bias, because the estimand itself is misidentified. The honest instrument here is not another variance damper but **sensitivity analysis** — quantifying how much unobserved confounding would be required to overturn a bridging verdict (e.g. Rosenbaum-style bounds or a confounding-strength parameter on $\hat{\pi}$) — reported as a robustness interval on B_{LCB} rather than a point claim. The exploration pool's randomized slice bounds this in the limit (randomized exposure is unconfounded by construction), but only for the fraction of traffic it covers. Named, bounded, not closed.
3. **Bridging is non-monotone in audience.** B_{LCB} certifies bridging over the *exposed* set; a post bridging at 10K may divide at 10M. The confidence must widen as the target population outruns the tested set — this ties back to saturation windowing and never fully closes.
4. **Low-dimensional opinion space** may under-represent true plurality; divisiveness along an unmodeled axis leaks into b_p. d is a real bias-variance knob (too small leaks divisiveness into support; too large smears the divide you care about).
5. **Peer-prediction collusion** and the high-precision clique (§10) — a structural equilibrium problem, not a bug.
6. **Recycling is mildly farmable** in principle (acting under-served), mitigated by model-estimated satisfaction but not eliminated.
7. **Distinguishing harmful from benign divides** (A's weighting) is a normative, instance-level choice with no purely technical answer.
8. **Sybil rings — mostly closed, with a residual** (§5, validated in C.5). A naive teleport-floor eigentrust let a single-target ring's *beneficiary* harvest the puppets' pooled floor mass (top-percentile influence as the ring grows). The shipped outgoing-diversity transmit weight closes this — a single-target puppet forwards nothing, and on RfA the ring collapses to the floor at every size with honest ranking preserved. The residual is theoretical: Cheng & Friedman prove no *symmetric* reputation function is Sybil-proof, so full resistance needs the *asymmetric* seeded teleport (supported, bound to the identity port) plus the forge-cost — and a sufficiently patient attacker who diversifies each puppet across many authors trades ring stealth for diluted impact. Hardened, not proven impossible.
9. B_{LCB} **now earns its complexity — where the data has group structure** (§4.2, validated in C.5). The earlier subtractive $\min_c$ penalty, fed raw rating counts, was beaten on Community Notes by both the scalar b_p and a naive mean (it subtracted noise, not risk). Replacing it with exposure-weighted empirical-Bayes shrinkage and the nash aggregator reaches b_p parity on CN and *beats* it on Polis's genuine multi-group divides (Spearman $0.80 \to 0.97$). Two residuals remain: against a label that is itself an intercept threshold (CN's CRH) a naive mean is unbeatable by construction, so CN is the wrong court of appeal for an aggregator; and the shrinkage is only as good as the exposure count n_{cp} feeding it — a rating-count proxy works, but a real propensity-corrected exposure model (§6) is the honest input.

10. **The camouflaged distributed ring — defended, with a residual** (§10, found *and fixed* in the simulator, App C.4). A ring that embeds puppets in *every* opinion cluster (by rating genuine content) and has them all boost one target fabricates cross-cluster support that beats B_{LCB}'s min *and* slips past the out-diversity λ weight (the puppets are not single-target raters); a plain co-approval discount is only partial (camouflage dilutes it). The attack's mechanism is a **rank-1 common-mode lift of the shared intercepts** b_p, b_a: the puppets scatter across clusters so their directional pull on y_p cancels, and only their shared positive residual survives, lifting every cluster's reception equally. The shipped defense keys on the one act camouflage cannot hide — the *same accounts* approving *every* one of the target's posts over time — and gates it by the loyal bloc's **opinion- cluster spread**, so a dispersed ring is penalized while a coherent single-cluster fanbase is not; in the simulator this drives the ring's amplification from 2.8× a legitimate author's reach back below 1×, even for a high-quality target. A second, complementary defense is also built — an *exploration-anchor cap* that bounds each cluster's reception by the unconfounded ϵ-exploration reception (§6.2); it *de-confounds* organic reception (raising delivered true value) and removes the ring's common-mode lift K-independently, but cannot *alone* push a ring below parity (a near-origin target's true reception equals genuine broad content's) and needs more randomized traffic than the floor provides — so it is paired with the loyalty penalty rather than used alone. *Spectral spike-removal* — deflating the rank-1 boost block from the residual — was also implemented and yielded an instructive **negative** result: per window, by reaction pattern alone, a coordinated ring is *indistinguishable from genuine cross-cluster consensus* (both are rank-1 residual blocks, and a genuinely-loved author's is the *larger*), so deflation suppresses the very bridging content the system exists to reward. The robust discriminator is therefore necessarily **temporal** (the loyalty signal — do the same accounts co-occur only on this target over time?) or out-of-band (identity/provenance); RPCA is doubly wrong (the boost lands in its low-rank part and is absorbed). Timing/provenance and costly identity remain the ultimate backstops, and this is the collusion analogue of the high-precision clique (#5).

Directions considered and declined. Several defensive add-ons were evaluated and left out deliberately, on the principle that each addition should strengthen an existing mechanism or sharpen a limitation rather than open a new subsystem: safe-policy (KL-to-chronological) regularization (redundant with the three variance controls above, and pulls toward a recency baseline the design rejects); stress-triggered L_2 boosting (raises convexity but flattens the very opinion geometry $D(p)$ depends on); and timing-entropy or clique-percolation collusion *detectors* (evadable by jitter, computationally heavy, and — critically on the fediverse — false-positive on legitimate tight-knit niche communities, which are the platform's atomic unit). The structural defenses of §10 (attacks made unprofitable) are preferred over evadable detectors. Adversarial-challenge auditing of suspicious bridged support is retained only as an anti-amateur measure; it does not defeat the §10 high-precision clique, whose sock-puppets already occupy the extreme coordinates a challenge would route to.

14. Relationship to prior work

Ordered roughly by how much of the working machinery each source contributes.

- **Bridging systems** [Ovadya & Thorburn 2023] — the conceptual foundation: the attention-allocation framing, the diverse-approval motif, and the validity/reliability cautions (notably that diverse *exposure* can backfire). Also the federated-deployment open problem this work takes up.

- **Collective response — Community Notes / Polis / YourView** — the **keystone**. The opinion-embedding matrix factorization that separates ideology from cross-cutting support (Community Notes / Polis), group-aware consensus and the clustering that B_{LCB} reconstructs over (Polis), and recursive cross-divide credibility (YourView). The single most important mechanism in the paper — the scoring keystone of §4 — originates here.
- **Community Notes / QSMF** — the deployed bridging MF at scale, its documented failure list (latency, surge instability, hyperactive-minority influence, manipulation) that validates our problem set, and the inverse-variance pathology plus the quality-tracking correction adopted in §5.
- **Counterfactual LTR** [Joachims–Swaminathan–Schnabel 2017; Schnabel et al. 2016; Steck 2010; Swaminathan–Joachims 2015; Dudík et al. 2014] — the MNAR/propensity core; everything in §6 that makes the keystone identifiable.
- **Performative prediction** [Perdomo et al. 2020; Brown–Hod–Kalemaj] — the outer-loop stability theory (§9).
- **Two-timescale stochastic approximation** [Borkar 1997; Lakshminarayanan–Bhatnagar 2017; Borkar–Pattathil 2018; Doan 2023] — the $\lambda \leftrightarrow$x convergence theory (§9).
- **Provider-side fairness / popularity bias** [Abdollahpouri et al.; Patro et al.] — the Matthew-effect / long-tail framing behind the budget and exploration mechanisms (§8), and the "re-ranker, not retrieval" architecture.
- **Multi-armed bandits** [Chapelle–Li; Li et al.; Dynamic Prior TS 2025] — the exploration pool and its base-rate-calibrated prior (§8).
- **Discoverers / "Hit-Savvy"** — empirical basis for scout precision (§5).
- **Ethelo** — the **objective** (strength = support net of dissonance), the **influence-recycling** mechanism (§8), and the framings of Rawlsian aggregation and constrained-outcome selection.

Appendix A. Notation

u user, p post, $a(p)$ author, c opinion cluster · r_{up} signed reaction · $x_u, y_p \in \mathbb{R}^d$ opinion embedding / loading · $\mu, b_u, b_{a(p)}, b_p$ biases · λ_u rater influence (quality-tracking) · $q_{\text{scout}}(u)$ scout precision · $D(p) = y_p^\top A y_p$ divisiveness · $A \succeq 0$ divide-weighting · $B_{\text{LCB}}(p)$ tested bridged support · π_{up} exposure propensity · ϵ exploration rate · $V(u,p)$ value · $\Phi(p)$ realized strength · $B(a)$ author budget · η replenishment · ζ recycling · knobs $M, \rho, \theta_f, \epsilon$.

Appendix B. One-line summary of the architecture

Rank by tested cross-cluster support net of weighted divisiveness; weight raters by quality-tracking not variance; price authors by a strength-replenished conserved budget; audition the unproven from a floored commons pool that doubles as the identifiability anchor; correct exposure MNAR with doubly-robust propensity weighting; stabilize the coupled estimator as two-timescale stochastic approximation held in a monitored bounded regime; and expose $M/\rho/\theta/\epsilon$ as knobs while keeping all authority earned.

Appendix C. Retroactive evaluation

No single public dataset exercises the whole system, and the feedback loop (§9) cannot be tested on any fixed dataset at all — a static archive cannot respond to the ranker's own allocations. Retroactive evaluation therefore validates the *static* components (scoring, rater weighting, ranking quality, debiasing) against public data, and defers the *dynamic* components (performative stability, exploration) to a simulator. The critical constraint is exposure: almost no organic social dataset logs what users were *shown*, so the propensity layer (§6) can only be validated on the rare datasets that carry a randomly-exposed holdout, or via a semi-synthetic harness where exposure is simulated so ground truth is known.

C.1 Layer-to-dataset mapping

Layer	What it tests	Best datasets	Notes
Keystone (§4): B_{LCB}, per-cluster reception, divisiveness	bridged support, opinion embeddings	**Community Notes** (rater×note ratings, public daily TSVs)	Same shape as the model; open-source CN baseline **and** QSMF baseline to benchmark against. Start here.
Rater weighting (§5): quality-tracking λ	manipulation resistance, extremist-amplification check	**Community Notes**	Reproduce the QSMF inverse-variance pathology and the correction directly.
Real deliberation / clusters	per-cluster reconstruction on genuine divides	**Polis** open conversation data	Small; participant×comment agree/disagree/pass with validated clusters.
Scale + real divides + authors + time	author budget, scout precision, ranking realism	**Reddit** (Academic Torrents dumps)	Votes not per-user; build x_u from community co-participation (Waller–Anderson).

Layer	What it tests	Best datasets	Notes
Propensity / MNAR (§6)	IPS / doubly-robust recovery	**Yahoo!R3, Coat, KuaiRec/KuaiRand**	The only datasets with a randomly-exposed (MAR) holdout = the unconfounded anchor. KuaiRand adds timestamps/features. **A true MAR block is essential, not a convenience** — see C.3: on Coat's real random-exposure test set IPW recovers the ranking, but running the same harness on a MovieLens slice (organically MNAR, no MAR block) it does *not*, because the "holdout" inherits the base selection bias.
Exploration / bandit (§8)	Thompson audition, base-rate prior	**Open Bandit Dataset**, KuaiRand	Logged uniform-random **and** Thompson-sampling policies.
Trust / credibility propagation (§5)	eigentrust λ on signed votes	**Wikipedia RfA**, Epinions/Slashdot signed nets (SNAP)	Rare per-user *signed* votes.
Temporal / author / scout only	budgets, early-signal backtest, ranking sandbox	**Hacker News**, StackExchange	No per-user votes $\rightarrow$ cannot do the keystone; use aggregate score as Φ_∞ and early *commenters* as a scout proxy.

C.2 Suggested sequence

1. **Community Notes** — implement B_{LCB}, per-cluster reconstruction, and the quality-tracking λ; benchmark against the open-source CN algorithm and QSMF. This proves the keystone and rater weighting against real baselines on real divides.
2. **Yahoo!R3 / Coat / KuaiRand + semi-synthetic (C.3)** — validate the propensity menu and the doubly-robust estimator against a known unbiased holdout.
3. **Reddit** — test author budget, scout precision, and end-to-end ranking on polarized topics at scale with real timestamps and authors.

4. **Simulator (C.4)** — the only way to exercise the closed loop.

C.3 Semi-synthetic propensity harness

Because the propensity model is deliberately an open menu (§6.3), the cleanest test gives you control of ground truth:

1. Take a near-complete or MAR ratings matrix R (e.g. the Yahoo!R3 / Coat / KuaiRand random-exposure block, or a dense MovieLens slice).
2. Define a synthetic logging policy $\pi_{up}^{\log}$ that exposes items in proportion to predicted in-group alignment — i.e. deliberately induce MNAR of the exact kind §6.1 warns about (in-group over-exposure).
3. Sample an observed set $E \sim \pi^{\log}$; hide the rest.
4. Fit the model on E under each propensity option (position/cascade, policy-derived, intervention-harvested from a held-in random ϵ-slice, doubly-robust wrapping).
5. Score against the *hidden* full matrix as ground truth. Report: (a) does uncorrected fitting inflate b_p / shrink $\|y_p\|$ (the predicted pathology); (b) does each propensity option recover the unbiased ranking; (c) how gracefully each degrades as $\pi^{\log}$ is misspecified relative to the fitted $\hat{\pi}$; (d) the value of the random ϵ-anchor — sweep its size toward zero and watch identifiability fail.

Metrics: unbiased NDCG@k / AUC against the MAR holdout (the standard for Yahoo!R3/Coat), plus the pathology diagnostics on b_p and $\|y_p\|$, plus an extremist-amplification curve for λ (reputation vs. ideological extremity, per QSMF Figure 1).

The base matrix in step 1 must be genuinely MAR (or near-complete) — this is a correctness condition, not a convenience, and it is easy to get wrong (Appendix C.5). The harness scores step 5 against a "hidden full matrix" taken as ground truth; if that matrix is itself the product of organic self-selection (as with a MovieLens or MovieLens-style slice, where people rate what they chose to watch), then the held-out cells carry the *same* unmodelled selection bias as the training cells. Correcting only the *injected* logging policy $\pi^{\log}$ then adds inverse-propensity variance without removing the pre-existing confound, and IPW can match or *underperform* the uncorrected fit — not because the correction is wrong but because the ground truth is contaminated. Empirically this is exactly what happens: on **Coat**, whose test block is a real uniformly-random exposure, IPW improves the unbiased ranking; run the identical harness on a dense **MovieLens** slice and it does not. Use the semi-synthetic recipe only on a dataset with a true random-exposure block (Coat, Yahoo!R3, KuaiRand) or a near-fully- observed matrix; treat any organically-sparse matrix as a negative control, not a validation.

C.4 What requires a simulator (not retroactive)

The feedback loop is untestable on fixed data. To exercise §9 you need an agent-based simulator with: a synthetic population placed in opinion space; a response model $P(\text{react} \mid x_u, y_p)$; content generation by author-agents that adapt to the incentive (to test whether the conserved budget actually suppresses firehosing and whether "bridging-bait" emerges); and the full closed loop (rank $\rightarrow$ simulated reactions $\rightarrow$ retrain). Targets to measure: convergence to a performatively stable point vs. oscillation as you vary the sensitivity knobs; the effective-rater-count / Gini controller (§9.3) holding concentration bounded; and whether exploration at rate ϵ sustains the identifiability anchor over time. This is where the performative-prediction and two-timescale results become empirical rather than assumed.

The reference implementation ships this simulator (`chord/simulator/`, see its `SIMULATOR.md`) built to three commitments that make it a real test rather than a flattering one: **counterfactual** — every ranker (CHORD, engagement, chronological, random, and a cheating oracle) runs its own closed loop on the same seeded world, so results read "what if we had ranked this way"; **non-circular** — the data-generating process is deliberately *not* the model the estimator fits (a hidden opinion axis beyond d; a toxicity channel that drives engagement while quality — genuine value — barely moves reactions, so engagement $\neq$ value and bridging-bait can exist); and **ground-truthed** — because the world is synthetic it scores what no live system can, the true bridged value delivered and the polarization exposed. The headline counterfactuals (seed-averaged): against an engagement ranker on the same world, CHORD delivers more true (quality $\times$ bridged) value and less toxicity/divisiveness at a modest satisfaction cost, and recovers the true opinion geometry better (its exploration anchor keeps the estimate identifiable in-loop); with strategic author-agents, engagement entrenches partisan extremity over time while CHORD does not; a sybil ring that boosts a target's posts *gains* reach under engagement but *loses* it under CHORD as the ring grows (B_LCB's min-over-clusters reads the ring as an outlier group); and the conserved budget (§8) dilutes a firehose author's per-post reach. One honest in-loop finding: the §5 out-diversity weight, which fixes the rater-influence ring, does not by itself stop this *content-boost* ring — the keystone does.

C.5 Results of the reference validation suite

The reference implementation ships an opt-in suite (`validate/`) that runs the static-component checks above against real public data — Coat, Polis, MovieLens-100K, Wikipedia RfA (SNAP), and a dense k-core slice of X Community Notes — with datasets committed via Git LFS. It is built to *surface* failures rather than confirm the design: claims that hold are assertions; claims that fail are recorded as documented findings, not tuned away. The first run is summarized here (indicative numbers, $d = 2$, averaged over seeds); several claims held and three did not.

Held. (i) On Coat's real MAR holdout, IPW correction improves the unbiased ranking (NDCG@5 $0.714 \to 0.718$, AUC $0.657 \to 0.663$) — modest but in the predicted direction (§6). (ii) On Polis, B_{LCB} tracks genuine cross-group support well (Spearman ≈ 0.71 against the minimum-across-groups mean vote), and the random-ϵ anchor demonstrably buys identifiability (unbiased NDCG falls monotonically as the anchor $\to 0$, §6.2). (iii) The §5 influence distribution is concentrated, not uniform ($N_{\mathrm{eff}}/n \approx 0.4$ on RfA).

Findings — claims that did not survive real data. (F1, §5) Teleport-floor eigentrust starves individual sybils but *not a ring*: a K-account ring pointing at one target lifts that target from the 75th percentile of real-editor influence at $K = 5$ to the 100th at $K = 100$ (the ring-harvesting analysis in §5). (F2, §4.2) On Community Notes, reconstructed B_{LCB} (AUC 0.93 vs. `CURRENTLY_RATED_HELPFUL`) is beaten by both the scalar b_p (0.95) and a naive mean of signed helpfulness (0.9994) — the pessimism penalty subtracts noise when n_{cp} is not a real exposure count. (F3, §6/C.3) The semi-synthetic IPW harness recovers the ranking on Coat but fails on an organically-MNAR MovieLens slice, because that slice has no true MAR block. Two weaker signals: Polis cluster reconstruction is strong on some conversations (vTaiwan ARI 0.59) and near-chance on others (football-concussions 0.04), and divisiveness $D(p)$ tracks group spread except on very low-conflict conversations.

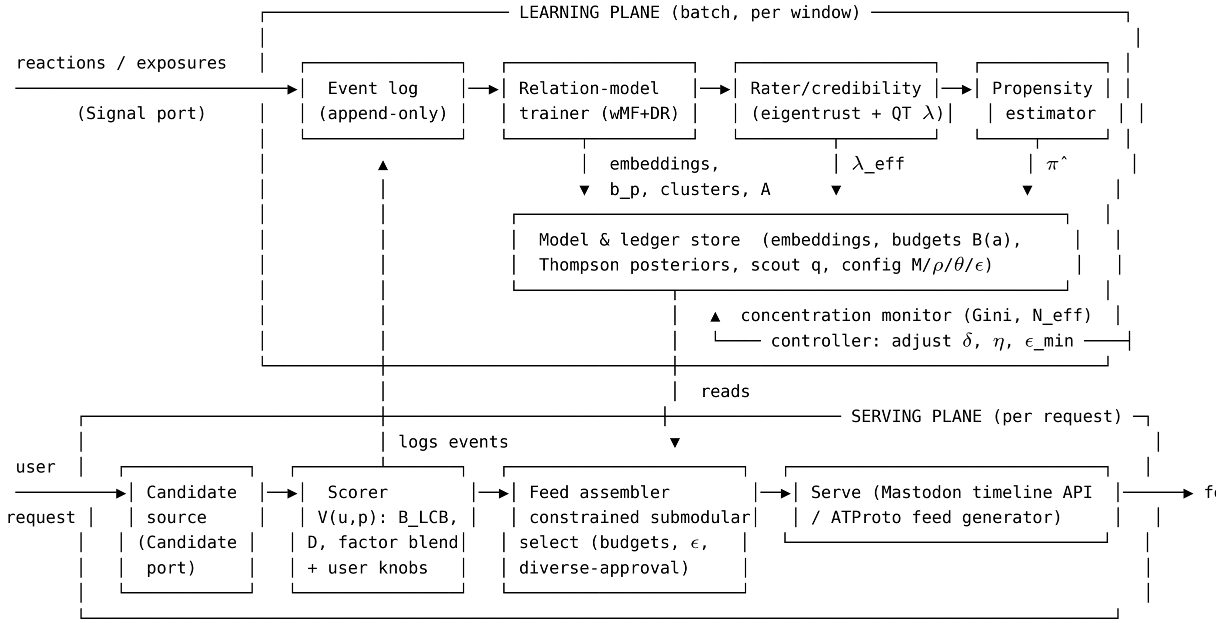
Both headline findings were then fixed, and the fixes are the shipped default. After a literature survey (trust-metric and welfare/risk/robust-statistics), candidate repairs were prototyped against the *same* benchmarks and promoted into the core. F1: the outgoing-diversity transmit weight (§5) collapses the RfA ring from the 100th percentile back to the 0th at every K, with honest ranking unchanged (Spearman 1.0). F2: exposure-weighted empirical-Bayes shrinkage plus the nash aggregator (§4.2) lift B_{LCB} to b_p parity on a balanced Community Notes sample and to Spearman 0.97 (from 0.80) against Polis's true cross-group support. The validation suite's F1/F2 tests, originally written to *document* the failures, now run green against the fixed code and guard against regression; F3 remains a standing caveat on the semi-synthetic recipe (use a genuinely-MAR base matrix), not a code defect. This find-it-then-fix-it loop — real data surfaces a failure, cross-disciplinary math supplies the repair, the same benchmark confirms it — is exactly what Appendix C is for.

Appendix D. Reference architecture for a fediverse deployment

A concrete sketch. The design is two planes — an offline **learning plane** (batch, per-window) and an online **serving plane** (per-request) — connected by an append-only event log and a set of model/ledger stores. Everything maps to the ports of §3; example technologies are illustrative, not prescriptive.

D.1 Deployment topology

ActivityPub/Mastodon has no native pluggable-feed-generator hook, so the ranker runs as an **instance-side re-ranking service** (a sidecar the instance's timeline endpoint calls) or as a **client-side ranker** consuming the user's home timeline via the Mastodon API. On Bluesky's AT Protocol the same service registers as a first-class **feed generator**, which is the cleaner integration and requires no instance cooperation. A crucial scale point: most instances are small (10^2–10^4 users), the ranker is a re-ranker over a few hundred candidates, and the MF trains on one instance's bounded reaction history — so this runs on modest hardware, which is precisely why it fits the fediverse rather than requiring hyperscale infrastructure.



Identity port (verified/ZK) and Preference port (portable pod: Solid / ATProto) sit alongside, read by both planes.

D.2 Components

Serving plane (low-latency, per request):

- **Candidate source** (Candidate port) — instance/home timeline, follow graph, or federated firehose. Retrieval stays upstream; the ranker only re-ranks.
- **Scorer** — loads x_u, and per-candidate b_p, y_p, cluster receptions, and A from the model store; computes B_{LCB}, $D(p)$, the factor blend, and $V(u,p)$ with the requesting user's knobs $M, \rho, \theta, \epsilon$.
- **Feed assembler** — greedy submodular constrained selection (author caps + budget $B(a)$, diverse-*approval* coverage, exploration floor ϵN). Optionally the constrained step is delegated to the AGPL Ethelo engine (Dockerized, JSON in/out) as a MINLP solver; a greedy $1-1/e$ selector is the lightweight default.
- **Serve + log** — returns the ordered feed via the Mastodon timeline API or an ATProto feed generator, and emits the exposure event (which slots were shown) plus subsequent reactions back into the event log. Logging **exposure**, not just reactions, is what makes the propensity layer possible — most systems fail to log it.

Learning plane (batch, per window):

- **Event log** — append-only stream of exposures and reactions (e.g. a Postgres table or a Kafka/Redis stream). The single source of truth.
- **Relation-model trainer** — weighted, doubly-robust MF (ALS); emits $x_u, y_p, b_u, b_{a(p)}, b_p$, the whitened divisiveness metric A, and opinion clusters (the default Partition-port adapter; a rich adapter can call an external Polis/bridging service).
- **Rater/credibility service** — damped eigentrust on the learned geometry + quality- tracking λ; applies influence recycling to emit λ^{eff}.
- **Propensity estimator** — pluggable (§6.3 menu), wrapped doubly-robust; calibrated against the exploration pool's known ϵ-exposures.
- **Ledgers** — per-author budgets $B(a)$ (strength-replenished), Thompson posteriors for audi-

tions, scout precision q_{scout}.
- **Concentration monitor / controller** — tracks effective rater count $(\sum \lambda)^2 / \sum \lambda^2$ and Gini(λ); if concentration climbs, raises the teleport floor δ, damping η, and $\epsilon_{\min}$. This is the §9.3 runtime guard that keeps the coupled estimator in its bounded regime.

The two planes run at different cadences — this **is** the two-timescale separation of §9: the serving plane and the fast MF refit are the fast timescale; the λ/credibility update is the slow timescale.

D.3 Cross-cutting ports

- **Identity** — verified-human / ZK-pseudonymous handle; the author budget and Sybil resistance bind here, not to raw accounts. Default adapter: account-age/heuristic forge-cost.
- **Preference** — the user's opinion embedding, factor knobs, and scout/credibility history live in a **portable data pod** (Solid or ATProto record) the user controls, so moving instances carries the profile. Default adapter: local store.
- **Config** — instance defaults for $M, \rho, \theta, \epsilon$ and the constraint set (a governance surface), with per-user overrides for the consumption knobs; earned quantities $(\lambda, q_{\text{scout}}, B(a))$ are never user-writable.

D.4 Federation notes

Clustering and the relation model are computed **per instance** on that instance's own reaction history — no central authority, and the default Partition adapter needs no external service. Cross-instance reputation flows through the trust graph (a graded alternative to binary defederation). Because the influent function is portable, a user's profile is not captured by any one instance. A maximal host that fills every port richly (verified/ZK identity, portable pods, external Polis clustering, a feed substrate, a governance UI) is an existence proof that all adapters can be provided, but the architecture runs on a single vanilla Mastodon instance with only the default adapters.